```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Titanic Dataset - Exploratory Data Analysis (EDA)

Objective

The objective of this project is to explore the Titanic dataset using statistical summaries and visualizations to identify patterns, trends, relationships, and anomalies.

```
import pandas as pd

df = pd.read_csv("tested.csv")
df.head()

df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  418 non-null    int64
 1   Survived     418 non-null    int64
 2   Pclass       418 non-null    int64
 3   Name         418 non-null    object
 4   Sex          418 non-null    object
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64
 7   Parch        418 non-null    int64
 8   Ticket       418 non-null    object
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     object
 11  Embarked     418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	418.000000	418.000000	418.000000	332.000000	418.000000	418.000000	417.000000
mean	1100.500000	0.363636	2.265550	30.272590	0.447368	0.392344	35.627188
std	120.810458	0.481622	0.841838	14.181209	0.896760	0.981429	55.907576
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000	0.000000
25%	996.250000	0.000000	1.000000	21.000000	0.000000	0.000000	7.895800
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000	14.454200
75%	1204.750000	1.000000	3.000000	39.000000	1.000000	0.000000	31.500000
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000	512.329200

Dataset Preview

The first five rows of the dataset are displayed to understand the structure and available columns.

```
df.isnull().sum()
```

```
df["Survived"].value_counts()
```

Survived	count
0	266
1	152

dtype: int64

Missing Values

The dataset is checked for missing values before data cleaning.

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Fare'] = df['Fare'].fillna(df['Fare'].mean())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

Data Cleaning

Missing values were handled by filling Age and Fare with the mean, Embarked with the mode, and removing the Cabin column because it contained many missing values.

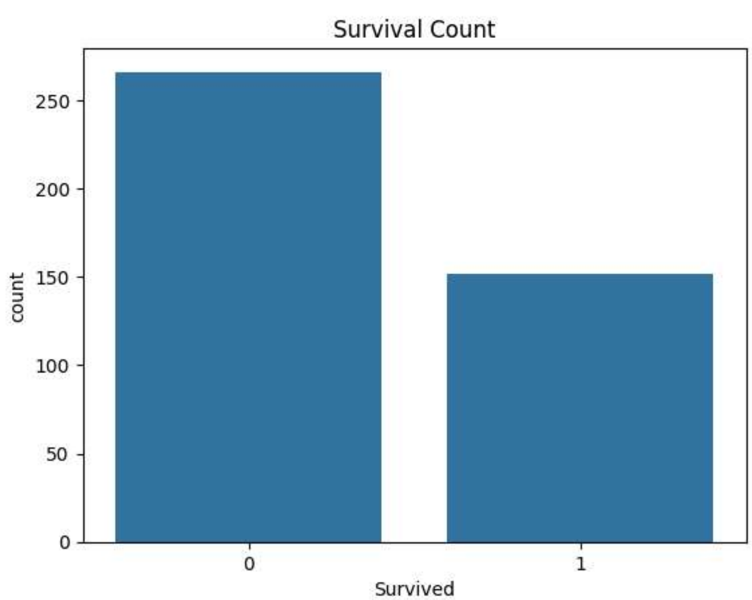
```
import seaborn as sns
import matplotlib.pyplot as plt
```

Import Libraries

Seaborn and Matplotlib libraries are imported to create visualizations for Exploratory Data Analysis (EDA).

Survival Count

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```

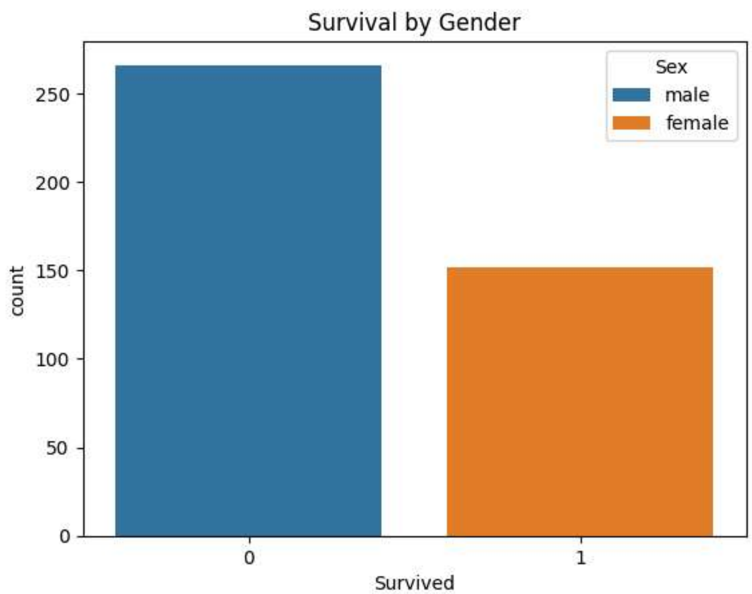


Observation

The number of passengers who did not survive is higher than those who survived.
This shows that the overall survival rate was low in the Titanic disaster.

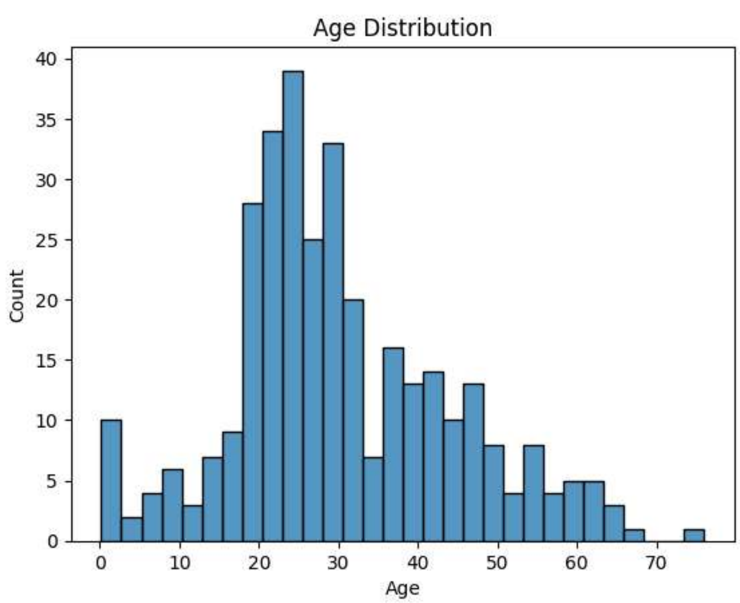
```
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Fare'] = df['Fare'].fillna(df['Fare'].mean())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival by Gender")
plt.show()
```



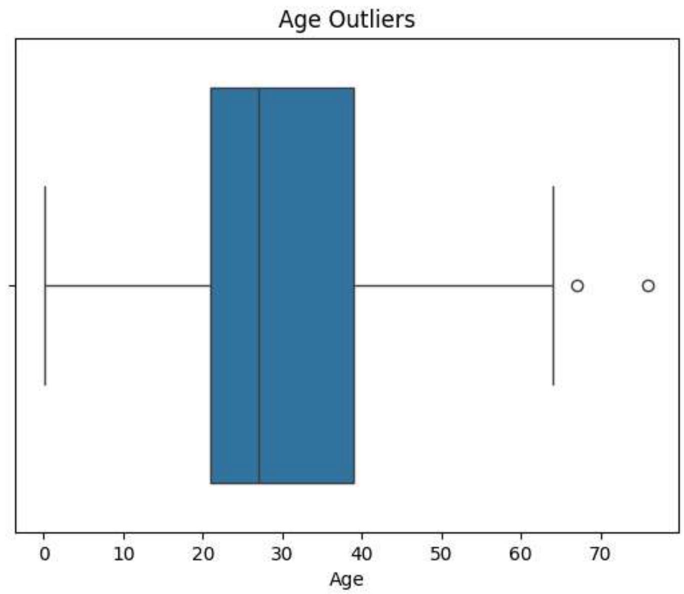
Females had higher survival rate compared to males. Most males did not survive, showing gender played an important role in survival.

```
sns.histplot(df['Age'], bins=30)
plt.title("Age Distribution")
plt.show()
```



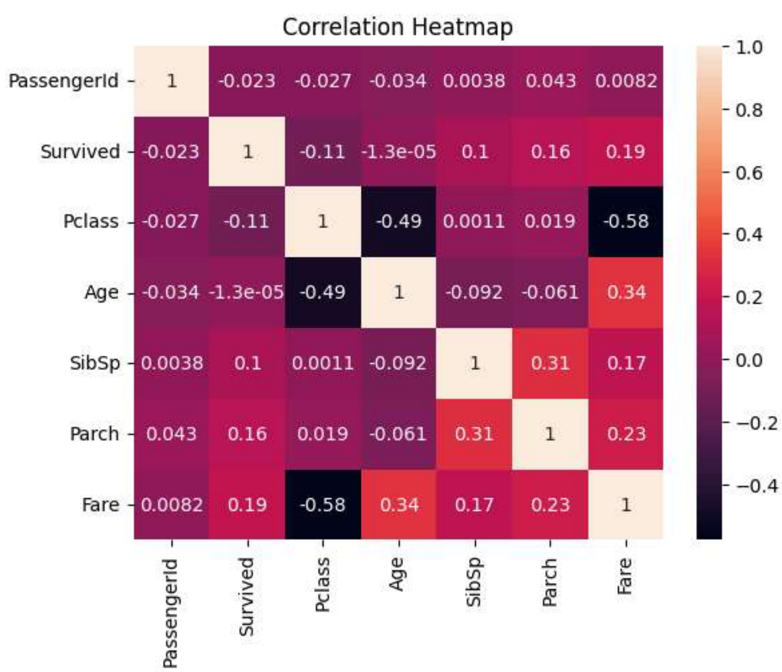
Most passengers were between 20 to 40 years old, showing that young adults formed the majority of the dataset. Very few elderly passengers were present.

```
sns.boxplot(x=df['Age'])
plt.title("Age Outliers")
plt.show()
```

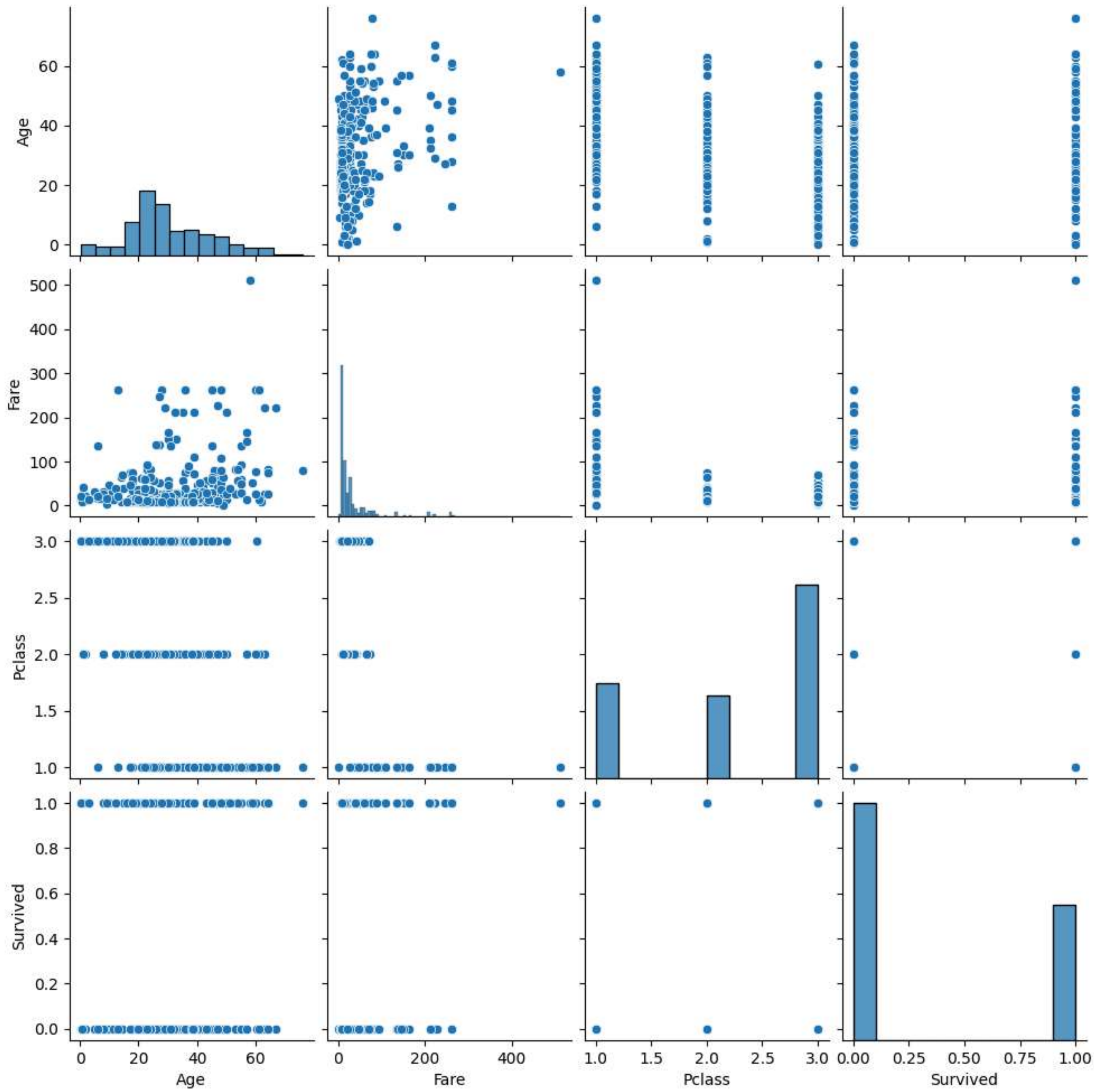


Age contains some outliers. Most passengers fall within a normal age range between 20 to 40 years.

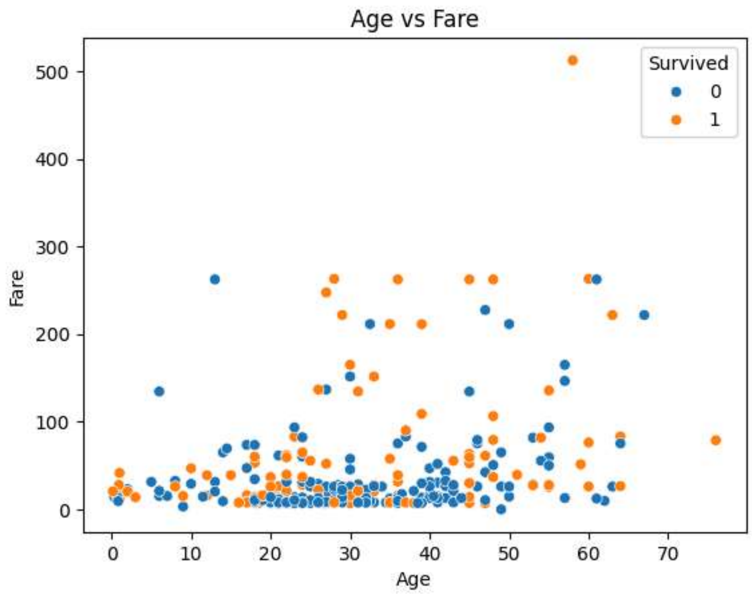
```
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```




```
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']])
plt.show()
```




```
sns.scatterplot(x="Age", y="Fare", hue="Survived", data=df)
plt.title("Age vs Fare")
plt.show()
```



Fare and Survived show a slight positive correlation. Pclass and Survived show a negative correlation, meaning lower class passengers had lower survival rate.

```
df.drop(columns=['Cabin'], inplace=True)
```